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Accuracy of farmer-generated yield estimations of common bean in decentralised on-farm trials in sub–Saharan Africa

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ABSTRACT

Improving agricultural productivity and resilience is essential to meet future food needs in sub-Saharan Africa under changing climate conditions. Achieving this will necessitate the development of high-yielding locally adapted crop varieties to mitigate the impacts of climate change. Despite advancements in crop improvement, varietal turnover in smallholder farms remains notably low. Continuous turnover of locally adapted varieties is essential, necessitating active dissemination of new varieties and withdrawal of obsolete ones across diverse target populations using participatory breeding approaches. A decentralised experimental approach, known as tricot, supported by citizen science, has proven effective in accelerating genotype selection while promoting inclusivity and diversity. However, the methodology has strongly relied on farmer-generated rankings, which provide relative performance insights but fall short in informing breeders with absolute yield data, limiting the ability to measure genetic gain or assess economic returns on breeding investments. To address this gap, we validated the accuracy of farmer-generated yield data for common bean (Phaseolus vulgaris L.), by comparing it with technician-generated volumes and researcher-generated absolute yield data. Results revealed strong correlations between farmer and technician volumes ($r = 0.96$, $p < 0.001$). The mean difference in farmer-technician log-yield was close to zero, indicating significant agreement. We further developed a predictive model to estimate absolute yields using farmer showing minimal influence from intrinsic and extrinsic factors. Our findings demonstrate that farmer-generated yield data can reliably inform breeding decisions and support the accelerated turnover of improved varieties. Integrating such data into breeding programs offers a cost-effective and scalable pathway to enhance agricultural productivity and sustainability across smallholder systems in sub-Saharan Africa.

1. Introduction

In sub-Saharan Africa (SSA), increasing agricultural productivity and ensuring food security remain pressing challenges, particularly in the face of rapid population growth and climate variability (Van Loon et al., 2019). To meet this demand, it is critical to develop high-yielding crop varieties with local adaptability under unpredictable climate conditions. Although significant strides have been made in the development of improved seed varieties in the Global South, their overall use and varietal turnover in smallholder farms remain significantly low (Spielman and Smale, 2017; Thiele et al., 2021). This can primarily be attributed to the fact that the evaluation and validation of improved varieties by breeding programmes has predominantly occurred in central locations that have the best soils and access to the best technologies, which is not representative of the changing and prevailing conditions on smallholder farms (Blum, 2018; Ceccarelli and Grando, 2007). Moreover, differences between centralised locations and farmers' fields are further exacerbated by other factors, such as weather, farming systems, and farmer skills (Marchant et al., 2019). To mitigate these challenges, there has been increased incorporation of on-farm experimentation and farmer participation in the evaluation of new technologies i.e., new improved varieties (Kool et al., 2020). These participatory approaches

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aim to increase the wider applicability of innovations (which in this case are new improved varieties) by farmers as well as increase their impact and validation in smallholder farming systems (Vanlauwe et al., 2019).

Increasing varietal turnover on smallholder farms would require participatory approaches for crop variety testing that address G x E (genotype by environment) interactions while incorporating traditional knowledge and considering site-specific sociocultural complexities (Fadda et al., 2020). Despite their significance, common traditional participatory approaches have major challenges because they are seldom streamlined or scaled due to their resource intensity in terms of time, training, and cost per beneficiary. As such, current approaches may not be adequate to maintain a continuous cycle of variety performance experimentation and selection of better-yielding genotypes under continuous climate variability, diverse environments, and management practices (De Sousa et al., 2024). To increase varietal turnover in smallholder farms, it would be necessary to implement breeding activities that deliver a steady stream of incrementally improved cultivars that not only excel in yield but also thrive in multiple target populations of environments (TPE) (Atlin et al., 2017; Atube et al., 2021; Mubiru et al., 2018).

A novel decentralised on-farm testing supported by citizen science, known as the triadic comparison of technology options (tricot), has emerged as a pioneering framework for on-farm testing. This decentralised participatory approach that engages a large number of farmers has demonstrated the ability to help breeders accelerate the selection of genotypes while addressing G x E x M interactions (genotype by environment by management) (de Sousa et al., 2021; Occelli et al., 2024; van Etten et al., 2019). Moreover, the approach promotes inclusivity by capturing diversity in regions where environmental conditions or sociocultural preferences vary strongly across the landscape. The approach also presents an opportunity where citizen science data can also be amalgamated with data on climate, preference, cropping systems and farming households for in-depth and meaningful analysis geared towards target-specific recommendations and strategies. Despite the great leaps achieved through the large-scale evaluation of varieties, the tricot approach still relies on farmer-generated rankings (Etten et al., 2019), which do not capture the performance of tested varieties on an absolute scale. According to Sousa et al. (2021), this aspect was a main hindrance to breeding programs that objected to capturing the performance of tested technology options on non-absolute scales. This would particularly make it difficult to measure genetic gain or translate trial results into monetary value terms, e.g., return on investment (ROI) from donor funds.

The challenge of having limited data available to measure heritability and selection gain in farmers' field conditions has persisted for decades (Soleri et al., 2000; Williamson and Leonelli, 2022). This has been attributed to the resource-intensity nature of data collection in terms of time, training, and cost per beneficiary. More importantly, farmer participation has always been hindered by institutional constraints and prejudices about the quality of farmer data (Morris and Bellon, 2004; Steinke et al., 2017). Although smallholder farmers have demonstrated proficiency in reporting accurate yield data (Diskin, 1997; Verma et al., 1988), these instances have occurred in yield surveys aimed at measuring productivity in smallholder production systems. Therefore, there is a need to demonstrate the accuracy of farmer-generated yield data in on-farm experimentation and its suitability for supporting breeding decisions. This is especially crucial in an era where public breeding objectives and donors stress the need to measure genetic gain within farmers' fields (De Sousa et al., 2024). This paper addresses this gap in research by experimentally assessing the accuracy of farmer-generated yield estimations in decentralised on-farm trials of common bean (*Phaseolus vulgaris* L.) in Tanzania. Furthermore, we also evaluate which intrinsic and extrinsic factors may limit the accuracy of farmer-generated yield estimations. In this case, the intrinsic factors included seed properties like moisture content and seed geometry factors, e.g., thickness, length, and width of seed, while extrinsic factors included age, gender, experience with extension services and farmer field schools, acuity, and literacy. By examining the suitability and reliability of metric data collection within the tricot framework, this study sought to clarify the potential of farmer-generated data in enhancing on-farm testing methodologies in supporting breeding decisions, an effort that would ultimately accelerate farmers' adoption of new improved crop varieties.

2. Materials and methods

2.1. On-farm trial implementation

The on-farm trials were designed and implemented in 2022 following the tricot approach, which is based on principles of decentralised on-farm testing involving randomly assigned varieties of common bean (*Phaseolus vulgaris* L.) being distributed to each farmer as incomplete blocks of three. A total of 355 farmers who participated in the tricot trials were selected based on gender, land availability, communication abilities and interest. The experiment was designed using the ClimMob platform (Quirós et al., 2024), accessible at https://ClimMob.net. Masking the variety names in ClimMob was done to anonymise their identity and avoid bias during evaluation. Each tricot package contained three varieties of bean varieties labelled with letters A, B, and C. Extension officers then distributed the trial packages to the farmers and registered each participant with identifiable data using the Open Data Kit (ODK). This data included the name, age, village, district, and GPS coordinates. Participants established the experiment on their own farms and assessed a list of traits as per trial protocol. In this case, the trial protocol was developed by the bean breeding program at TARI (Tanzania Agricultural Research Institute) and PABRA (Pan African Bean Research Alliance) based on the common bean target product profile (TPP) for Tanzania and standardized using Crop Ontology terms (https://cropontology.org, Guerrero et al., 2023).

All varieties used in the tricot trials were phenotypically distinct and uniform (Table 1). Both men and women farmers planted the three varieties in sub-plots of 6 × 8 m next to their own crop under usual crop management for the common bean. Ultimately, each tricot experiment followed the balanced incomplete block design comprising three blocks, ensuring A-optimality by connecting all the varieties to each other in the complete set (de Sousa et al., 2024). Farmers evaluated the varieties using a simple ranking method by answering two short questions for each trait performance aspect. For example, for yield, the questions would be "Which variety had the highest yield?" and "Which variety had the lowest yield?". Farmers then communicated their field observations to the extension officer verbally, on paper, or via mobile phone. Independence of observations was ensured by not revealing the names of varieties and asking participants individually for their results without bias from external parties.

2.2. Yield determination survey

The yield determination survey was conducted in 2022 with 30 farmers (24 men and 6 women) from Kilindi and Karatu districts in the Northern zone of Tanzania. The annual temperature in Karatu district averaged 18°C with a bimodal rainfall pattern. Karatu typically receives approximately 254 mm in December, while July is the driest month, receiving about 29 mm of rain. On the other hand, Kilindi district has a semi-arid climate with an average annual rainfall ranging from 400 mm to 700 mm and temperatures ranging between 22°C and 30°C throughout the year. In each district, the yield determination protocol was provided and explained to farmers and individuals present in the household before the survey, after which they would find the best person to take on the exercise, preferably the person who would normally do this kind of exercise (Appendix I). On harvest day, each farmer was interviewed for 60–120 minutes using the yield survey questionnaire after securing consent to be interviewed. The interviews were conducted

Table 1
Variety names, market class, key traits, potential yield, testing and release years of common bean varieties evaluated in tricot trials in the Northern Tanzania during the 2022 crop season.

Name of variety	Market class	Key traits	Potential yield in tonnes	Testing year	Release year
RCB593	red	Heat tolerant	1.2	-	2021
SCR61	red	Heat tolerant	1.3	-	2021
CODMLB033	red mottled	High yielding	1.4	-	2021
KAB−06F2–8–36	red mottled	Iron and zinc	1.5	-	2021
RWR2154	sugar	Iron and zinc	1.6	-	2021
SMC18	white	Tolerant Anthracnose and Angular leaf spot	1.7	2016	Breeding line
Sweet violet	sugar	Tolerant to Angular leaf spot, rust, bacterial blight	1.8	2016	Breeding line
Gloria	sugar	Tolerant to Common bean rust and bacterial blight	1.9	2016	Breeding line
Kablanketi	kablanket	Early maturing	1.10	2016	Breeding line
NUA 48	red mottled	Tolerant to Angular leaf spot and Anthracnose	1.11	2016	Breeding line
NUA 64	red mottled	Tolerant to Anthracnose	1.12	2016	Breeding line
Kipapi	kablanket	High yielding	1.13	2016	Breeding line
VTT923/23–10	sugar	Tolerant to Common bean rust	1.14	2016	Breeding line
KATB9 (check)	red	Early maturing	1.15	-	2018
Uyole03 (check)	sugar	Cooked fast	1.16	-	2003
Selian 12	red	Early maturing	1.17	-	2018
Uyole 03	sugar	Cooked fast	1.18	-	2003

in five sections: farmer characteristics and demographics, farmer acuity and literacy, farmer-led and researcher-weight measurements, and plot-related parameters (Appendix II). Data on farmer characteristics, acuity, and literacy were collected to determine their effects on the accuracy of farmer yield estimates. Farmer acuity was measured using Snellen's charts with farmers standing 6 m away from the evaluator.

There were three types of yield collected in the study, namely farmer-led estimations of yield volumes, technician-led estimations of yield volumes, and researcher-weighted yield. For the farmer-led yield measurements, farmers were provided with a version of the yield determination protocol two weeks before harvest (during a field visit or by phone) and before the interview. The protocol, printed with visual aids for better understanding, was administered and thoroughly explained by the /researcher, after which the participants were free to withdraw from the exercise. Farmers were encouraged to harvest the three plots separately on the same date and dry the beans in their pods for two days in their respective plots. Afterwards, farmers were to shell the pods and leave the separated grains in the sun for another two days (Fig. 1A). Farmers were also asked to record each plot's harvest dates and the number of surviving plants. On the day of the yield determination survey, farmers were provided with 1 L of *sado* tins to measure the volume of each variety (Fig. 1B).

Farmers and technicians measured yield using tins to mimic real on-farm stations where most farmers cannot afford or access the tools to measure absolute yield in metric units. The *sado* tins were specifically used for this study because of their ubiquity and commonality in most parts of Tanzania. The farmer reported the number of tins for each variety at the end of the exercise, which the technician recorded in the ODK. Afterwards, the technician carried out the same protocol for measuring yield volumes following the same protocol provided to the farmers (Fig. 1C). Both farmers and extension officers were separated during yield estimation to prevent bias. Farmers and extension officers were advised to take volumetric measurements on a flat surface and level the tins using a large flat object (more significant than the tin diameter) to ensure that the tins were not heaped. The farmer or extension officer counted and reported the number of tins for each variety at the end of the exercise. Additional data collected in this section included losses during the growing season and after harvest, visual estimation of yield in the field and harvest heaps and heaping versus levelling of tins.

Metric measurements were also collected by the researcher supervising the work. We recorded moisture content (%) and seed weight (kg) after the exercise on yield volume estimation. For moisture content, three moisture readings were taken for three separate samples of each bean variety using a moisture meter (Twist-Grain pro model) (Fig. 1D). Other collected measurements included a hundred-grain weight and seed geometry (length, width, and thickness). A hundred seed weight was determined by measuring the weight of 100 seeds of each variety (Fig. 1D). Readings were taken three times using different samples of the same variety. The thickness, length, and width of ten seeds from each variety were measured using a vernier caliper for seed geometry.



Fig. 1. A: Farmer in Karatu district shelling the pods of the common bean varieties to prepare for yield determination. B: Farmer from Kilindi district, with two containers: a 1 L sado tin (blue) given to her by the researcher and the white mug that she usually uses to measure seeds. C: An extension officer taking volumetric measurements of yield. D: Researcher taking moisture content readings of the varieties in the tricot trial. Other measurements included yield in grams, hundred seed weight, and seed geometry measurements (seed thickness, length, and width).

2.3. Data analysis

All yield survey data, including volumetric measurements by farmers

and metric measurements by the extension officer (technician), was imported in R (v4.2.2) (R Core Team, 2024). The data and code used in this analysis is available on Zenodo (Nabateregga et al. 2025). The data was then combined with data sets from ClimMob using tricot package IDs. The Pearson correlation coefficient, also known as the bivariate correlation, was used to determine the strength of the linear relationship between farmer and researcher volume estimates (Pearson, 1895). Additionally, we assess whether the medians of farmer-generated yield data and technician data were significantly different. Firstly, the Shapiro-Wilk test assessed the normality assumption of the pair differences between grain yield and volume estimations. This assumption was rejected. Therefore, the Wilcoxon sing-ranks test was used to evaluate whether the median of the differences between paired observations is significantly different from zero (Wilcoxon, 1992). To further assess accuracy of farmer-generated estimates, the Bland and Altman (1986) method of agreement were performed. This is a statistical method for assessing agreement between two methods by quantitatively estimating the closeness between farmer and technician volume estimates using log-scale.

We aimed to assess whether intrinsic and extrinsic factors influence the accurate determination of yield by farmers in smallholder settings. To achieve this, we use the following linear mixed-effects model, which we named as the 'full model':

$$Yield = FV + variety * (MC + T + W + L) + age + gender + ESE + FFS + AT1 + AT2 + AT3 + LT + \underline{farm} + \underline{error} \quad (1)$$

Where "Yield" represents the researcher-weighted yield in kilograms, FV denotes the farmer volume in tins, and "variety" refers to the seed variety used. The intrinsic factors include MC, moisture content in percentage, T, thickness, W, width, and L, length of the seeds, with T, L, and W measured in millimetres. The extrinsic factors are related to the farmer and include age, gender, EES (experience with extension services), EFFS (experience with farmer field schools), EMV (experience in measuring volumes), acuity as assessed by the scores obtained in three tests (AT1, AT2, and AT3), LT, the score from the literacy test. For the remaining terms, "farm" represents the farm effect, and "error" refers to the plot-level residual, representing non-systematic variation due to within-field heterogeneity and experimental error. The "* " symbol indicates that both main effects and interactions are considered. The underlined predictors are considered random effects, while the remaining predictors are treated as fixed effects.

The influence of the intrinsic and extrinsic factors is analysed according to their significancy to explain the variation of yield. This significancy is assessed using a feature selection approach based on a step-down strategy (Diggle, 2002; Schwarz, 2009). In this process, multiple nested models were evaluated sequentially, starting with the full model presented in Eq. (1). Through this step-down procedure, non-significant variables are systematically removed based on F-test, which leads to identifying the most relevant predictors for yield. This method ensures that only the most relevant factors remain in the final model, reducing model complexity while maintaining predictive accuracy. The goodness-of-fit of the final model is evaluated by both the marginal and conditional coefficients of determination (R_c^2 and R_M^2, respectively). The marginal R_M^2 represents the variance explained by only the fixed effects, while the conditional R_C^2 reflects the variance explained by both fixed and random effects. For this analysis, the Acuate Test 3 (AT3) and Literacy Test (LT) effects were removed from the dataset because they had the same values across all farms. Additionally, 13 observations were discarded, 3 of them due to having a value of 0 for farmer volume, and 10 because they contained at least one missing value. All fixed effects with a quantitative scale were centred before fitting the model. The random effect *farm* was always retained, as it is considered relevant for the correct specification of the model. The significance of the remaining effects was tested.

Furthermore, a regression analysis was used to determine the predictive model for absolute yield at plot level across the tricot trials using farmer-generated volumes. The study employed a linear mixed-effects model, treating farmer-reported volume as a fixed effect and farm as a random effect to capture unobserved variability at the farm level. Yield was measured per plot, with three plots per farm. Cross-validation, conducted using 3-folds and 100 repetitions, was performed at the farm level to ensure realistic model evaluation. During validation, predictions were based only on fixed effects, excluding random effect estimates, since these would not be available for farms that were not part of the training dataset. This approach supported the model's practical applicability and its intended use for generalization across diverse on-farm contexts.

Finally, we applied a simulation approach to ascertain the number of farmers required to generate volumetric measurements for the best likelihood of acquiring accurate results. The idea is to define the minimum number of data points required for data calibration and also a minimum set of field visits that would be required by trial implementers to monitor data quality. In this context, the sample size was the number of farms used in the training dataset during each iteration of the simulation. The study tested different sample sizes (e.g., 5, 10, 15, 20, etc.) by randomly extracting a subset of farms from the simulated dataset of 1000 farms. Each selected sample was used to train the predictive model, and the process was replicated 100 times for each sample size to evaluate its effect on model accuracy. The predictions are generated using only the fixed effects of the model, as they are generated for new farms not included in the original dataset.

The procedure is performed as follows:

1. A simulated dataset of 1000 farms/experiments is generated using the predictive model for absolute yield, which was estimated from the original data.
2. From the 1000 simulated farms, 100 are randomly selected for validation, and 5 are randomly selected for training.
3. The farms selected for training are used to estimate the predictive model for absolute yield.
4. The predictive model obtained in Step 3 is then used to predict grain yield for the farms selected for validation in Step 2.
5. The predicted grain yield from Step 4 is compared with the actual values from the validation set, and the Root Mean Square Error (RMSE) and the Coefficient of Determination (R^2) are calculated.
6. Steps 2–5 are repeated for different sample sizes of the training data (10, 15, 20, 25, …) and the process is replicated 100 times for each sample size.

3. Results

3.1. Accuracy of farmer-led yield data

The mean volumetric yield of common bean varieties in the farmer-generated and technician-generated approaches were 1.13 tins and 1.26

Table 2
Mean, median and standard deviation of researcher-weighted yield (kg per plot), farmer volumes of yield and technician yield estimates collected from Karatu and Kilindi districts in the Northern zone of Tanzania.

Yield estimation approach	Researcher weighted yield (kg)	Farmer-generated volume (number of tins)	Technician volume (number of tins)
Maximum	3.45	3.75	4.50
Minimum	0.05	0.00	0.00
Mean	1.00	1.13	1.26
Median	0.73	0.75	0.90
Standard deviation	0.80	0.93	1.03

tins, respectively (each from 6 ×8 m plots) (Table 2). Farmer-generated volume measurements varied widely, with a median of 0.75 tins, while technician-measured volumes had a slightly higher median of 0.90 tins (Table 2). Researcher-weighted yield as measured by the researcher ranged from a minimum of 0.05 kg to a maximum of 3.45 kg, with a mean yield of 1.00 kg and a standard deviation of 0.80 kg (Table 2).

In terms of crop performance, Sweet Violet was the highest-yielding variety for both farmers and technicians, while RCB 593 had the least yield, as measured by both groups (Fig. 2). Gloria and NUA48 varieties were not captured in the sample for the yield survey and, as such, do not have means (Fig. 2). Also, farmers were able to correctly estimate volumes with a few tendencies to under-estimate and/ or over-estimate yield (Fig. 3A). Nevertheless, our study generated approximately 90 data points, of which only three instances of misreporting were reported due to the resulting low grain yield volume.

A comparative analysis of correlation coefficients between farmer-generated volume, technician volumes and researcher-weighted yield showed significantly strong linear correlations between all variables (Fig. 3A). We found that grain volume estimated by technicians had a strong correlation with researcher-weighted yield values in weight ($r = 0.991$), which was the highest accuracy (Fig. 3A). Moreover, there was also a strong correlation between farmer-generated volume and technician volume at $r = 0.96$ (p-value < 0) (Fig. 3B). High correlation values between farmer and technician volumes are indicative of a high accuracy of farmer-generated estimates of yield.

The Bland-Altman test showed that the mean difference in log-yield (farmer–technician) was close to zero, which confirmed agreement between technician and farmer yield volume (Fig. 4). Values falling outside the mean + 0.5 and mean −0.5 SD indicate a mismatch in volumetric yield caused by the farmer measurements (Fig. 3), which, as explained before had the tendency to underestimate the yield volume.

3.2. Effects of intrinsic and extrinsic factors on the accuracy of yield estimates

Our study also evaluated the effects of intrinsic and extrinsic factors on the accuracy of farmer-generated yield estimations. Intrinsic factors included moisture content and seed geometry factors, e.g., thickness, length, and width of seed. Regression analysis on the intrinsic effects of moisture content on yield estimations revealed a significantly low correlation coefficient ($R^2 = 0.119$) (Fig. 5). This implied that moisture content explained only 11.9 % of the differences observed in the density and weight of common bean varieties. These findings indicate that the farmers correctly applied the yield determination protocol which ensured minimal variability in moisture content.

The analysis of the relationship between grain yield with variable moisture content and dry yield across genotypes showed a strong positive linear relationship between these two variables (Fig. 6). The findings suggested that genotypes with higher grain yield tended to have higher dry yield (Fig. 6). As such, there was consistency in yield ranking, though the actual yield values shifted based on moisture adjustment. Importantly, our findings also showed that the spread of moisture content (ranging from 14.5 % to 16.0 %) did not appear to disrupt the positive association, though some of the larger-sized points (higher moisture content) were slightly below the fitted line. This implied that there was a tendency for overestimation of grain yield at higher moisture levels, which became corrected when yields were expressed on a dry matter basis.

Feature selection applied to the fixed effects resulted in a reduced model containing six predictors, as presented in Table 3.

The reduced model explains a considerable proportion of yield variation, with $R^2_M = 0.923$ and $R^2_c = 0.956$. Only intrinsic factors were significant. To assess their contribution to the goodness-of-fit, we fitted a simpler model excluding intrinsic predictors, retaining only farmer volume (FV) and variety, which yielded $R^2_M = 0.898$ and $R^2_c = 0.912$, indicating a loss of performance of less than a 5 %. A further model including only FV produced similar results $R^2_M = 0.897$ and $R^2_c = 0.906$. As variety and intrinsic effects provided minimal gains in goodness-of-fit, they were omitted from the predictive model.

3.3. Predictive model of yield of farmer-generated volumes of common bean yield

Building on the previous analyses and validations, we developed a predictive model to enable the conversion of farmers' volumetric estimations into grain yield. Although we expected market class (a classification in common bean based on seed size and colour) to be an important factor in predicting grain yield, a likelihood ratio test suggested that this effect was not significant ($\chi^2 = 3.73$, df = 3, p = 0.292). Market class was initially considered in the predictive model based on the assumption that differences in grain size, shape, and density across market classes could influence the relationship between volumetric estimates and actual grain weight. A high coefficient of determination indicated that seed volume was the main variable causing the proportion of variation in the seed weight. Overall, the results show that farmer volume estimates explain a large proportion of the variation in grain yield even without special consideration of variations in market classes.

The final model is represented by Eq. (2):

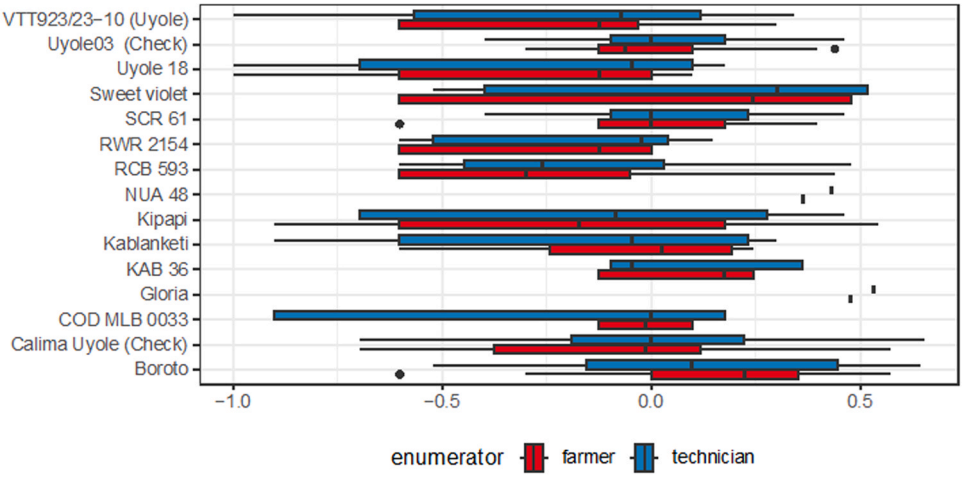


Fig. 2. Distribution of farmer-generated and technician-generated volumes of yield based on number of tins (sado tin). The figure shows common bean varieties evaluated in the tricot on-farm trial network in the Karatu and Kilindi districts in the northern zone of Tanzania.

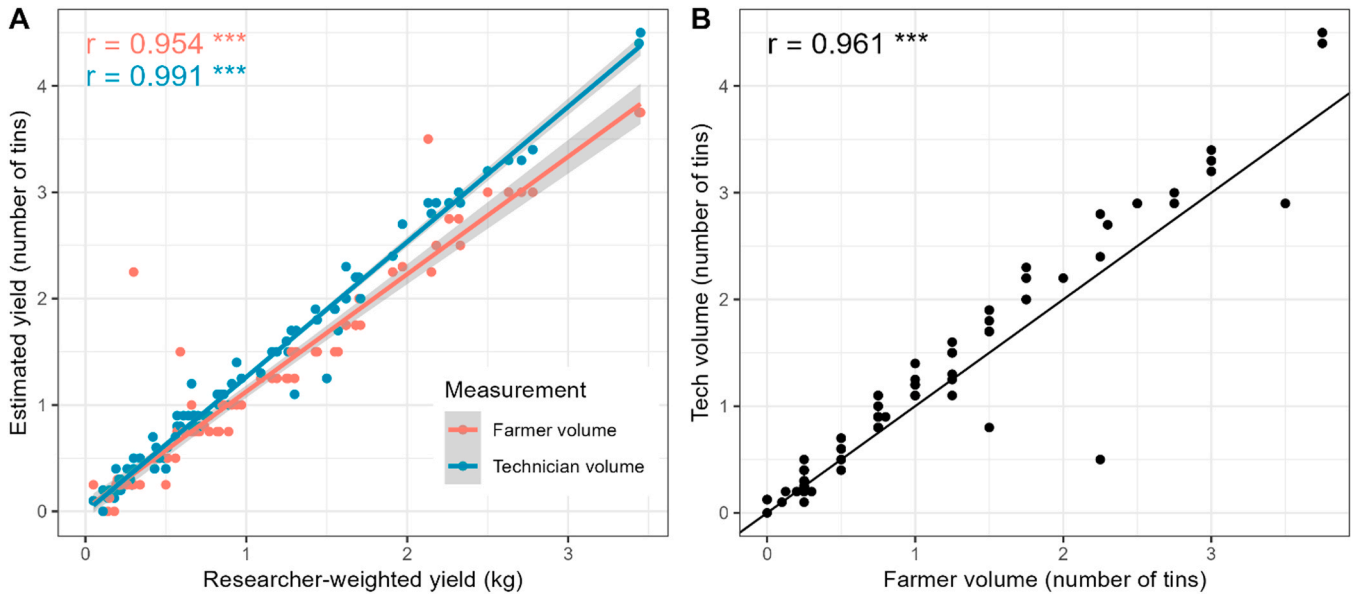


Fig. 3. Relationship between estimated yield from different sources. (A) Correlation between researcher-weighted grain yield, farmer-generated volumes, and technician-generated volumes. (B) Correlation between technician and farmer generated volumes. Pearson's correlation coefficient (r). The black line indicates the 1:1 relationship (complete correspondence). * ** Level of significance p value < 0.

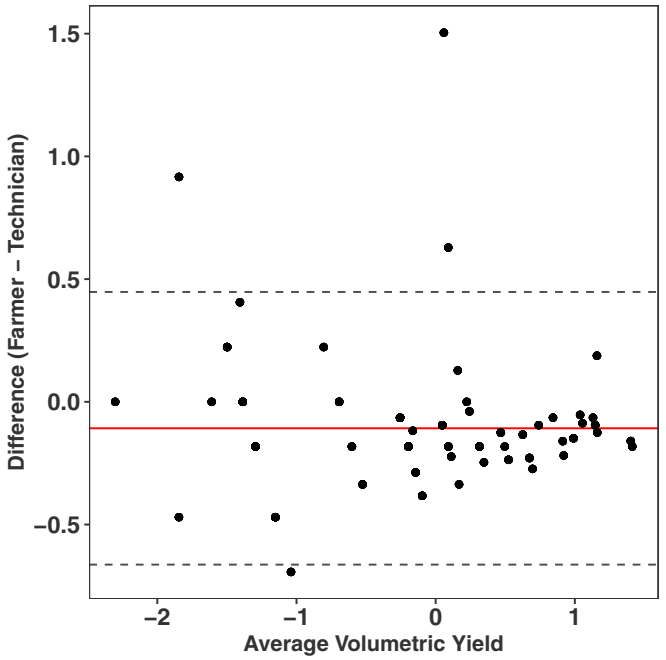


Fig. 4. Differences between volumetric yields collected by farmers and technicians. Values are in log-scale. The solid red line represents the mean difference in log-yield (farmer–technician), the upper dashed lines represent the mean + 2 SD (standard deviation), and the bottom dashed line represents the mean−2 SD.

$$Yield = FV + \underline{farm} + \underline{error} \quad (2)$$

Where, farmer volume in tins (FV) has a fixed intercept and a fixed slope, while the random effects 'farm' and 'error' are assumed to be normally distributed with mean 0 and standard deviations σ_{farm} and σ_{error} respectively.

We assess the predictive performance of the model specified in Eq.(2) using cross-validation with three folds and 100 repetitions. The metrics that were analysed included the root mean square error (RMSE), coefficient of determination (R^2), and mean absolute error (MAE) (Table 4). The low RMSE value (0.233) suggested that the model's predictions deviated from the actual grain yield by approximately 0.233 units. Additionally, the high R^2 value (0.901) implied that the model explained a substantial proportion (approximately 90.1 %) of the variability in the observed grain yield (Table 4). We also observed a low MAE value (0.133), which represented the absolute difference between the predicted and observed grain yields, further supporting the accuracy of the model. Overall, the model based on farmer volumetric estimations shows strong predictive performance, validating farmer-generated data in aiding breeding decisions for specific target environments.

3.3.1. *Final model*

An analysis of the estimated fixed effects and variance components showed that farmer volumetric estimates had a significant positive effect on grain yield, as indicated by the high level of statistical significance of the estimated coefficient (B1 = 0.884, p value < 0) (Table 5). This implied that higher farmer volumetric estimates are associated with increased grain yield, with each unit increase in farmer volumetric estimates corresponding to an increase of approximately 0.884 ton $\cdot$ ha^{-1} in grain yield in grain yield. The intercept term (B0) was 0.043, with a standard error of 0.021, indicating an expected yield of approximately 0.043 tons per hectare when all predictors are zero. The variance component was associated with random effects (0.049 and 0.094) (Table 5). These findings suggested that there were sources of variability in grain yield beyond the influence of the explanatory variables included in the model. These differences could be attributed to differences in location and climatic conditions between the Kilindi and Karatu districts.

3.4. *Simulations for the number of participants*

The results on the simulation approach to ascertain the number of farms required showed that the predictive model performs well in terms of RMSE and R^2 across different sample sizes, even with a low number of farms i.e., around 10 and 20 (Fig. 7). The distribution of the RMSE values decreases and stabilises when the number of farmers is 25, however, its RMSE values varies only between 0.09 and 0.14 kg. Likewise, the R^2, which is a goodness-of-fit measure for linear regression models, was substantially high for all participation levels from 5 to 100

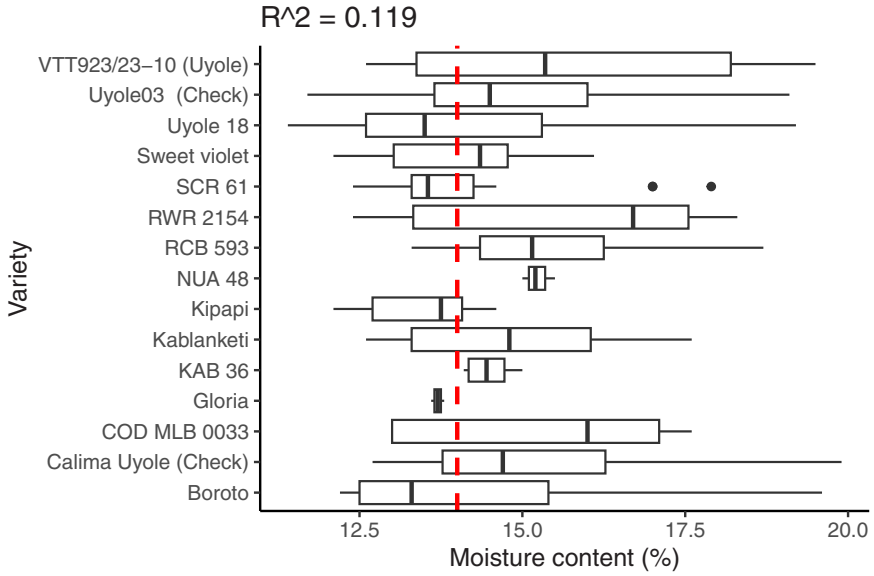


Fig. 5. Variation in seed moisture content (%) across different bean varieties at the time of farmer sampling. The red dashed line represents 14 % moisture content, the reference level at which common bean varieties are evaluated at research stations.

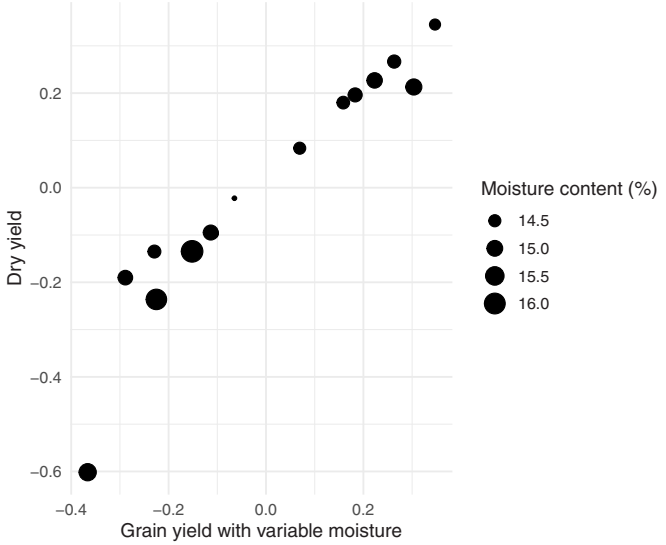


Fig. 6. Relationship between grain yield with variable moisture content and dry yield across genotypes. The scatterplot illustrates the correlation between grain yield (measured at varying moisture contents) and standardized dry yield. Each point represents a genotype, with the size of the point proportional to its moisture content (%), ranging from 14.5 % to 16.0 %. Larger points indicate higher moisture levels.

farmers (between 0.98 and 0.99; Fig. 6). These results indicate the robustness of farmer-generated volumetric measurements even with a small number of farmers contributing with data.

4. Discussion

Our study demonstrated that farmer-led estimations of yield could differentiate between common bean genotypes. Although previous studies reported contradicting performances of varieties like Sweet Violet (Amongi et al., 2023; Beebe et al., 2007) and RCB 593 (Chaves-Barrantes et al., 2018; Polania et al., 2016), these inconsistencies can be owed to G x E interactions an aspect that is critical for farmer adoption of new varieties. Our findings suggested that the farmer-generated volumes can capture the crop performance of common bean varieties, an opportunity that would significantly reduce the cost of

Table 4
Evaluation metrics of the predictive performance of the mixed model for predicting grain yield from farmer volumetric estimations: cross-validation with 3 folds and 100 repetitions.

	Mean	IC_Linf	IC_Lsup	max	min
RMSE	0.233	0.223	0.243	0.080	0.434
R2	0.901	0.890	0.912	0.995	0.425
MAE	0.133	0.129	0.136	0.065	0.223

RMSE is the root mean square deviation, R2 is the coefficient of determination, MAE is the mean absolute error, IC_Linf is the lower confidence interval, and IC_Lsup is the upper confidence interval.

Table 3
Type III analysis of variance table evaluating the significance of intrinsic and extrinsic factors on the estimation. The effects included were selected through a feature selection process applied to the fixed effects.

Source	SS	MS	Ndf	Ddf	F value	P-value	
Farmer volume (FV)	11.652	11.652	1	27.550	387.819	> 0.001	* **
variety	0.637	0.046	14	24.583	1.514	0.179	
Thickness (T)	0.074	0.074	1	24.649	2.474	0.128	
Width (W)	0.014	0.014	1	32.564	0.461	0.502	
variety:T	0.847	0.085	10	25.728	2.818	0.017	*
variety:W	1.305	0.109	12	25.873	3.619	0.003	* *

Significance codes: 0 '* ** ' 0.001 '* *' 0.01 '* ' 0.05 '.' 0.1 ' ' 1. Source: The fixed effects and their interactions, SS: Sum of squares, MS: Mean sum of squares, Ndf: Numerator degrees of freedom, Ddf: Denominator degrees of freedom, calculated using Satterthwaite's approximation, F: F-statistic and p-value: p-value.

Table 5
Estimated fixed effects and variance components of grain yield (ton ·ha^{-1}) with farmer volumetric estimates, model Eq. 1.

Farmer volume (FV)		
Intercept	0.043	*
Standard error	0.021	
Slope	0.884	* **
Standard error	0.014	
Random effects		
σ_{farm}	0.049	
σ_{error}	0.094	

σ_{farm} and σ_{error} represent the standard deviations of the farm-level and residual random effects, respectively. Significance codes: 0 * **, 0.001 * *, 0.01 *, 0.05 +

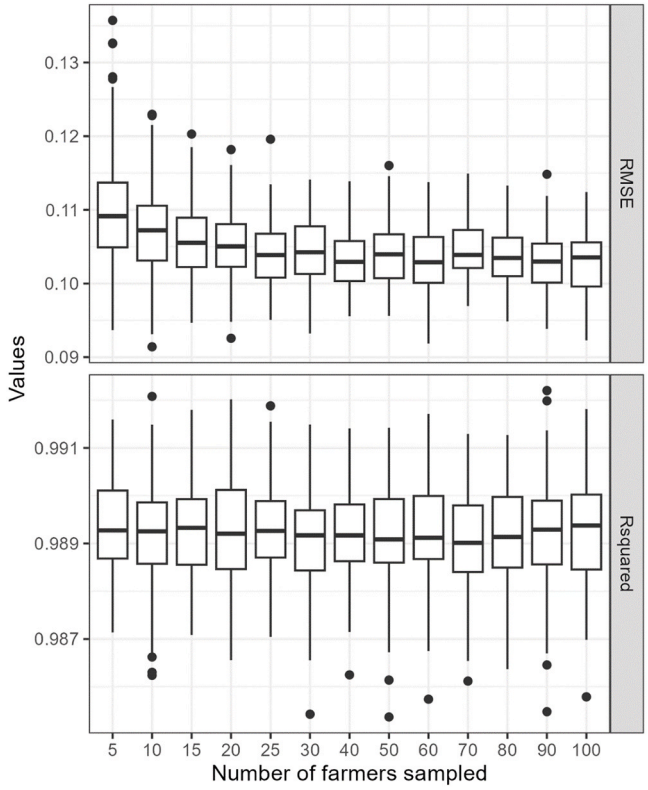


Fig. 7. Simulation analysis for different numbers of participating farmers (5−100). The median values and percentiles of 1000 runs are shown.

on-farm evaluation when performed on a large scale. However, it is important to note that this study used a non-random sample from a larger trial, and therefore, the yield levels reported were not representative of overall variety performance.

Most importantly, our findings from Pearson's correlations, the Wilcoxin test, and the Bland-Altman test of agreement highlighted the accuracy and robustness of farmer-generated estimates as equivalent to technician-generated estimates in on-farm experimentation. The choice of common bean as the study crop may have influenced the accuracy of farmers' yield estimates due to its relatively larger seed size and well-established harvesting methods. Particularly, high correlation values between farmer and technician volumes are indicative that both well-instructed farmers and technicians can generate accurate yield estimations from on-farm plots. Ochieng et al. (2019) reported similar findings with significant and strong correlation between the researcher's grain yield and the farmer's volumetric grain yield ($r = 0.97$ and p-value < 0.001) in participatory on-farm trials aimed at evaluating the adaptability of selected common beans in western Kenya.

Despite the significant accuracy of farmer volumes, we found that there were a few tendencies to over-estimate and/or under-estimate small fractions. According to Murphy et al. (1991) and Ochieng et al. (2019), farmers tend to avoid reporting yield estimates in fractions sometimes with the argument that these volumes would not be considered in the market. The variation in farmers' observations could also be explained by differing views on "good measures" among participating farmers, i.e., heaped *versus* levelled tins (Fermont and Benson, 2011). This same study also reported the farmer might assign a higher rating to a variety with a greater market value compared to one with a lower market value due to the anticipated income. In our study, the low seed volumes could have presented challenges in measuring fractions by farmers. Although the common assumption is that farmer-generated estimates are the only source of errors, the researcher's method of measuring grain yield may also be exposed to biases. Sources of bias may be related to weighing problems, such as the use of inappropriate weighing scales (large scale intervals), faulty weighing scales, and taring container weights. As much as individual errors may be small, a combination of errors can be significant (Murphy et al., 1991; Ochieng et al., 2019). These instances were not observed in our current study, but future experiments should consider the above sources of bias.

Our evaluation of the effects of intrinsic seed factors on yield volume showed that moisture content had no significant impact on the weight and density of seed. Primarily, these findings indicated that a large portion of farmers correctly applied the yield determination protocol which ensured minimal variability in moisture content. Previous studies have highlighted significant deviations in grain weight due to variations in moisture content. For example, Casley and Kumar (1989) reported significant differences of up to 100 % between regions and 70 % within regions in Nigeria. Another study by Murphy et al. (1991) revealed that moisture content of fresh maize at harvest time was as high as 20 or 25 % and about 10 % at the point of storage. However, it should be noted that the sole purpose of these studies was to report spatial crop productivity and not for experimental purposes. Moreover, farmers harvested seed from large plots compared to the small plots used in our study.

It is also important to note that seed was collected from standardized plots of 6 × 8 m in our study, which generally provided lower seed volumes that are easier to dry. The suggestion to increase plot size would be impractical, as larger plots would place an additional burden on farmers and increase the risk of measurement errors due to inconsistent drying. As such, using small plots was advantageous in this regard, especially for smallholder farmers in sub-Saharan Africa, where farm sizes are typically less than 0.5 ha and resources are limited (Giller et al., 2021; Ritzema et al., 2017). Another important factor to consider is the issue of crop area estimation. According to Nabateregga and Sølberg (2024), inaccuracies in crop area estimation can lead to significant errors in yield calculations as yield is often expressed as a function of area. This is especially crucial in smallholder settings where plots are often fragmented and irregularly shaped, making crop area estimation challenging.

Extrinsic factors that included age, gender, interaction with extension officers, farmer field schools and experience in measuring volumes, had no effect on farmer-generated yield accuracy. Although gender was found to have no significant effect on yield estimation accuracy, the small number of female farmers (six) participating in our study may have limited the ability to accurately assess gender-related differences. This imbalance could have affected the findings and highlights the need for more inclusive sampling in future studies. Similarly, while participation in farmer field schools (FFSs) did not significantly influence accuracy, previous studies have emphasized the role of farmer organizations in fostering participation and knowledge exchange (Danial et al., 2007; Greenberg, 2018). Contrary to our findings, Ulzen et al. (2024) found that farmer accuracy in heap volume estimation of maize kernels and cocoa beans was influenced by factors such as

location, eyesight, education, and experience. However, tin-can measurements in this study were largely unaffected by these variables. Differences between our common bean study and the study conducted by Ulzen et al. (2024) on maize kernels and cocoa beans may stem from variations in crop characteristics, study design, farmer demographics, and context-specific farming practices.

Ultimately, our findings reinforced that farmer-generated volumes were sufficiently accurate to predict on-farm metric yields. Although our hypothesis was that market class would be an important factor in predicting grain yield, a likelihood ratio test suggested that this effect was not significant ($p = 0.292$). Our sampling exercise, indicate that even with data from as few as 25 farmers, trial managers can still benefit from highly accurate farmer-generated yield data. This finding has important implications for scaling on-farm testing trials as highlighted by de Sousa et al. (2024). It enables cost-effective expansion by reducing the need for extensive field visits traditionally required in technician-generated yield data. Overall, the results show that farmer-generated volumes can reliably predict yield under real farming conditions even without accounting for grain size variations across different marked classes.

5. Conclusion

Our study, for the first time, demonstrates that farmer-generated yield estimates, when supported by proper protocols, offer a robust, cost-effective, and scalable alternative for evaluating crop performance in on-farm trials. Despite longstanding concerns regarding the reliability of farmer-generated data, our results reinforce that farmers can generate yield data that are highly consistent with technician measurements when standardized protocols are applied. Nonetheless, it should be noted that our study particularly focused on common bean, a crop with larger seeds and standardized harvesting methods that facilitate estimation. Estimating yields for smaller-seeded crops like wheat, sorghum or millet could be more challenging due to differences in seed size, threshing methods, and moisture content variability. As such, further research is needed to assess the reliability of participatory yield estimation methods across different crop types that may require larger volumes to be harvested. Another crucial consideration would be the estimation of crop area, especially in smallholder systems where plot shapes and sizes are often irregular and fragmented. Furthermore, a few farmers' training investments would be helpful to maximise and maintain the accuracy of farmer-generated yield estimates.

Based on the accuracy of farmer volume, our study developed a robust predictive model for converting farmer-generated volumes into metric yield, which indicated that farmer-generated volumes are effective predictors of grain yield even across varieties with differing seed sizes. Moreover, simulation further validated the robustness of farmer estimates across varying sample sizes, with accuracy expected to improve through repeated participation over multiple crop cycles. This has practical implications for scaling participatory trials while managing logistical constraints. Nevertheless, attrition rates must be accounted for, i.e., participants who drop out from the tricot experiment before successfully ending the trial. In conclusion, our study confirms that farmer-generated yield estimates are accurate, scalable, and well-suited for decentralized breeding efforts geared towards food security. With proper guidance and standardization, farmers can play a pivotal role in on-farm variety evaluation, contributing valuable, cost-effective data for breeding programs and policy decision-making.

CRediT authorship contribution statement

Theresia Gregory: Resources, Methodology, Investigation. **Kauê de Sousa:** Writing – review & editing, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joost van Heerwaarden:** Formal analysis, Investigation, Methodology, Supervision, Validation. **Mabel Nabateregga:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Hugo Dorado-Betancourt:** Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Svein Ø Solberg:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Jacob van Etten:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Ethics approval

The research received approval from the Institutional Review Board (IRB) of the Alliance of Bioversity International and CIAT (Appendix 5).

Consent to participate

Farmers provide consent to participate. All co-authors of this manuscript have consented to participate in this study and have their contributions published in the review.

Authors' contributions

KDS, JVE, SØS and MN conceptualised the paper. JVE was responsible for funding acquisition. MN, KDS, JVE and SØS were responsible for the methodology. MN and TG performed fieldwork and data collection. KDS, MN and HD were responsible for data curation. HD, JvH, KDS and MN performed data analysis. JvH and HD developed the data workflow to assess yield on-farm. KDS, JvH supervised the work of MN and HD. KDS, MN, HD, and SØS wrote the first draft. All authors read, reviewed, and approved the final manuscript.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.eja.2025.127730.

Data availability

Data and code are available on Zenodo (Nabateregga et al. 2025, https://doi.org/10.5281/zenodo.15591106).

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